Calibration of A TLD Personnel Monitoring Badge and Dose Evaluation

Date of experiment: 16 January 2026

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Date of submission: April 4, 2026

Contents

1 Objective **1**

2 Apparatus **1**
2.1 Specifications of The Irradiation Setup . . . 1
2.2 Badge Reader Specification . . . 1
2.3 Badge Reader Circuit Diagram . . . 2

3 Theory **2**
3.1 Detection Material . . . 2
3.1.1 Core Components and Geometry . . . 2
3.1.2 Filters in TLD Personnel Monitoring Badge . . . 3
3.2 Basic Structure and Traps . . . 4
3.2.1 Basic Insulator Band Model . . . 4
3.2.2 Traps and Luminescent Centers . . . 4
3.2.3 Trap Population and Filling . . . 4
3.2.4 Kinetic Model and Glow Curve Formalism . . . 5
3.2.5 Fading . . . 5
3.3 Heating, Recombination, and Light Emission . . . 5
3.3.1 Controlled Heating / Thermal Stimulation . . . 5
3.3.2 Migration and Recombination . . . 5
3.3.3 Glow Curve Peaks . . . 5
3.3.4 Example and Stability of Peaks . . . 5
3.4 Annealing . . . 6
3.5 Working Principle of a TLD Personnel Monitoring Badge Reader . . . 6
3.6 Reason for Using Three TLD Phosphors in a Personnel Monitoring Badge . . . 6
3.7 Dose, Temperature, and Energy Dependence of TL Intensity . . . 7
3.8 Advantages of Thermoluminescent Dosimeters (TLD) . . . 7
3.9 Limitations of Thermoluminescent Dosimeters (TLD) . . . 8
3.10 Heating Mechanism of the TLD Badge Reader . . . 8
3.11 Suggested Image Placement in Theory Section . . . 8

4 Observation **9**
4.1 Observation table of Data . . . 9

5 Calculation **10**
5.1 Error Calculation . . . 10

6 Conclusion **10**

1 Objective

- To do the calibration of a TLD personnel monitoring badge reader
- Checking the linearity of the calibration curve
- Checking the repeatability of the process
- Finding unknown dose of a TLD personnel monitoring badge

2 Apparatus

The following equipment and instruments were utilized in this experiment:

- TLD personnel monitoring badge
- Automatic TLD badge reader system
- Computer interface or software for reader control and data acquisition
- Radiation Source (X-Ray unit)
- Nitrogen gas supply (as required for the heating system)

2.1 Specifications of The Irradiation Setup

The irradiation was performed using a medical X-ray collimator. The detailed specifications are listed below:

Table 1: Specifications for Medical X-Ray Collimator SR-102

Field	Value
Product Name	Medical X-Ray Collimator
Model	SR-102
Product Part No.	2111904064702317
Input Power	AC 24V / DC 24V
Inherent Filtration	1.0mmAl
Max Tube Voltage	150kV
Production Date	2019.5.5

2.2 Badge Reader Specification

The readout of the TLD badges was conducted using an automatic badge reader system. Its technical specifications are as follows:

Table 2: TLD Badge Reader Specifications

Parameter	Details
Dosimeter	Three-element BARC $CaSO_4$(Dy) PTFE disc dosimeter badge with personnel ID.
Light Measurement System	Photo-Multiplier tube (R 6094) Light measuring system (LMS).
Dark current	Dark current is 1 μSv ($CaSO_4$) equivalent with software-based sampling & subtraction.
Heater Element	Nichrome heater assembly.
Heating Method	Hot gas (N_2) heating.
Nitrogen flow rate measurement	Digital flow rate meter.
Heating Cycle	The temperature is raised to 280°C in 8 s and clamped at 285°C.
Dose Range	50 μSv – 10 Sv (Gamma) and 100 μSv – 10 Sv (Beta).
Dose Threshold	$< 50\mu$Sv.
Readout time	Approx 100 s per badge.

Table 3: TLD Badge Reader Specifications (Continued)

Parameter	**Details**
Residual Signal	$\leq$ 10% of reading.
Facilities Available	Entry of calibration factor, etc. Storage of dose and glow curve data of badges in Hard Disk. Stepper motor driver assembly for automatic feeding of 50 dosimeter cards loaded in a magazine.
Software	Windows 7 (32 bit) / XP-SP3 compatible software facilitates storage and display of glow curves, computations of dose & generation of dose reports, transfer of data to user defined file, etc.
Temperature Monitoring	Chromel-Alumel thermocouple in hot gas stream.
Range Selection	Automatic.
Calibration	Coarse adjustment by varying the EHT through a potentiometer in the EHT circuit.
Safeguards	Heater / Gas flow / EHT / Card ID failure: The heater temperature, gas flow rate, card ID and EHT are checked for failure in every dosimeter readout cycle. In the event of failure of heater or gas flow rate or EHT or card ID, the readout is terminated and a message indicating corresponding failure message is flashed on the PC monitor.
Mechanical Failure and failure message	Any mechanical failure during readout cycle is sensed online & cycle is terminated with an option for the user to restart the cycle. EHT or the input circuit (I-F converter) is sensed and reading cycle is terminated in case of failure.
Nominal Power Supply	Power supply: 230 V, 50 Hz: Power requirements: 500 VA (max including PC).

2.3 Badge Reader Circuit Diagram

3 Theory

The thermoluminescent dosimeter (TLD) was first introduced in 1954 by Professor Farrington Daniels at the University of Wisconsin-Madison. When exposed to ionizing radiation, electrons within the TLD material are excited and subsequently captured by dosimetric traps. These electrons remain localized in these traps until the material is subjected to thermal stimulation. As the temperature of the detector increases—typically reaching up to 400°C—the trapped electrons gain enough thermal energy to escape. Upon returning to their ground state, they release energy in the form of a light pulse, a phenomenon known as luminescence. A photomultiplier tube (PMT) detects and quantifies this emitted light, the intensity of which is directly proportional to the radiation dose absorbed by the dosimeter [1, 2].

Due to its reliability, the TLD serves as a passive solid-state radiation detector and is widely recognized as a standard tool for personnel monitoring [3].

In the context of the Indian nationwide personnel monitoring program, $CaSO_4$:Dy is the preferred TLD material. This phosphor is capable of detecting X-rays, gamma (γ), and beta (β) radiations. Although it is not inherently tissue-equivalent, its adoption is justified by several significant advantages, including high thermoluminescent sensitivity, cost-effectiveness, ease of fabrication, and a straightforward annealing process. The concentration of Dysprosium (Dy) dopant in the calcium sulfate matrix is maintained at 0.05 mole %, equivalent to 500 ppm [4, 5].

From a clinical dosimetry standpoint, optimal TLD use requires controlled reader stability, robust calibration factors, uncertainty tracking, and periodic quality assurance checks. These operational principles are aligned with modern recommendations for luminescent dosimetry in medical physics practice [3].

3.1 Detection Material

3.1.1 Core Components and Geometry

A typical TLD badge comprises several key structural elements:

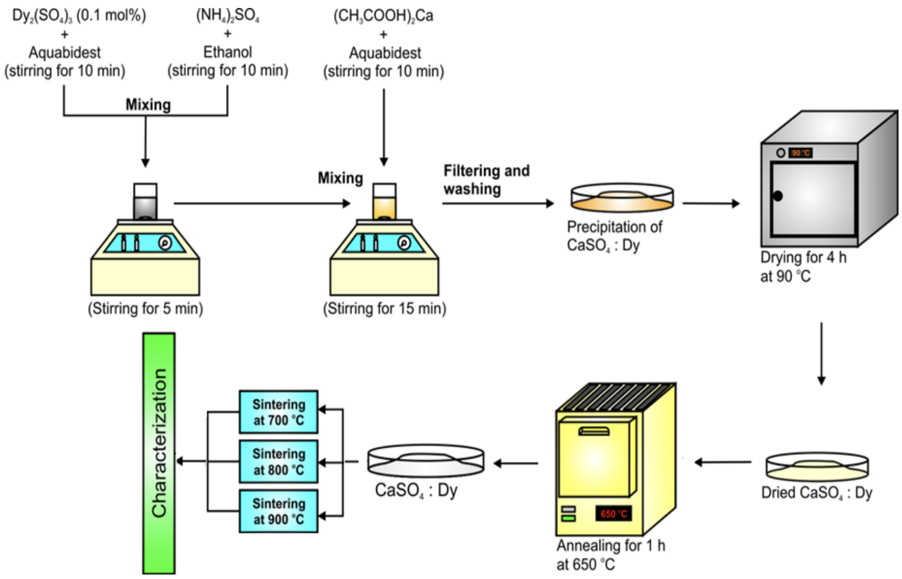

Figure 1: Schematic representation of co-precipitation method for synthesizing CaSO4:Dy.

- **Plastic Cassette/Holder:** This serves as the external casing, housing the TLD card and the associated filters, and offering physical protection to the internal components.
- **TLD Card:** Usually constructed from nickel-plated aluminum, this card supports the thermoluminescent material. It features specific dimensions (commonly 52.5 mm × 30 mm × 1 mm) and incorporates an asymmetrical shape or a "V" notch to guarantee proper alignment when inserted into the TLD reader.
- **TLD Discs/Elements:** The badge generally contains three or four small circular discs securely clipped into designated slots on the aluminum card.
  - **Material Composition:** Frequently employed TLD materials include Lithium Fluoride (such as LiF:Mg,Ti or LiF:Mg,Cu,P), Calcium Sulphate ($CaSO_4$:Dy or $CaSO_4$:Tm), and Aluminium Oxide (Al_2O_3:C). Inherently insulators, these base materials are doped with specific activator ions (e.g., Mg, Ti, Dy, Tm, or Cu) that create necessary trapping and luminescence centers within the crystal lattice.
    - LiF:Mg,Ti is highly valued for its tissue-equivalence, making it ideal for personnel dosimetry.
    - $CaSO_4$:Dy exhibits exceptional sensitivity but is characterized by a strong energy dependence.
  - **Dimensions:** The standard size for these discs is approximately 13.3 mm in diameter and 0.8 mm in thickness.

In practical low-dose X-ray fields, both $CaSO_4$:Dy and LiF:Mg,Cu,P show useful linear response and reproducibility, but $CaSO_4$:Dy generally provides higher sensitivity and lower detection threshold, which is particularly advantageous for personnel monitoring close to the dose limit [4].

3.1.2 Filters in TLD Personnel Monitoring Badge

To differentiate between various radiation types and energies, and to facilitate the accurate evaluation of both shallow (skin) and deep (whole body) doses, the TLD badge utilizes a three-filter system positioned over the individual TLD discs:

- **Metal Filter:** Comprising a combination of 1 mm of Aluminum and 0.9 mm of Copper, this filter mitigates the significant photon energy dependence of the underlying TL material.
- **Plastic Filter:** Designed to be tissue-equivalent, this filter mimics human tissue. It partially absorbs lower-energy radiation, thereby assisting in the estimation of the shallow dose imparted to the skin.

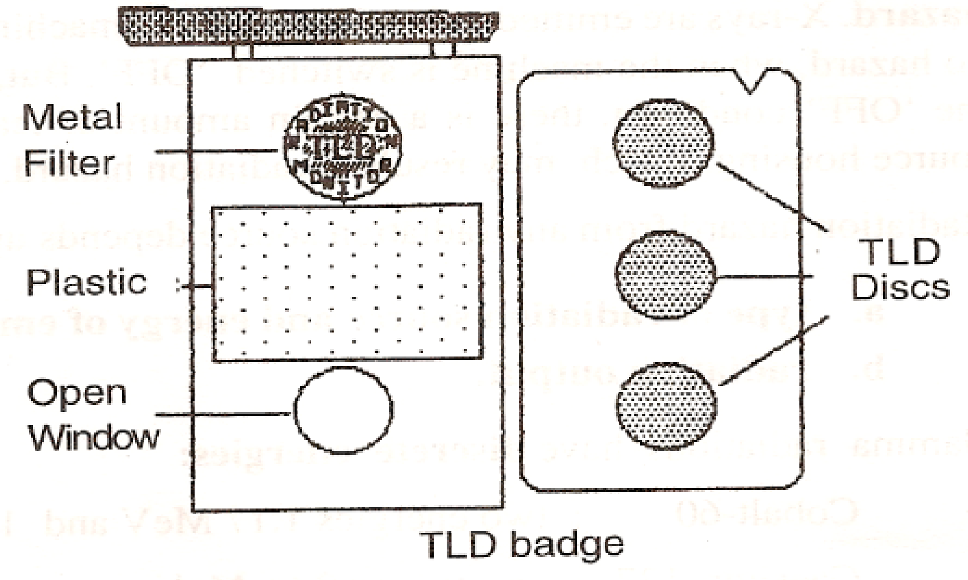

Figure 2: Schematic of a TLD personnel monitoring badge showing the arrangement of filters and TLD discs.

- **Open Window Filter:** This section lacks any covering, exposing the underlying TLD disc directly to the environment. It permits the detection of low-energy X-rays, gamma rays, and beta particles, and is primarily utilized for surface dose measurement and detecting beta contamination.

3.2 Basic Structure and Traps

3.2.1 Basic Insulator Band Model

TLD materials are fundamentally crystalline insulators or wide-bandgap semiconductors (with a bandgap exceeding 3 eV). In an ideal crystal, there are no energy levels within the forbidden gap. However, practical crystals contain structural imperfections, impurities, and lattice distortions. These defects establish localized energy states within the bandgap, functioning as either electron traps or recombination (luminescent) centers for charge carriers.

3.2.2 Traps and Luminescent Centers

- **Traps:** These are metastable localized energy levels within the forbidden band where charge carriers (electrons or holes) can be temporarily isolated, delaying their immediate recombination.
- The characteristic depth of a trap (its energy difference from the conduction band) governs the required thermal activation energy for carrier release.
  - **Shallow Traps:** These require minimal thermal energy to release electrons. They are easily emptied at lower temperatures and correspond to the early peaks observed in a glow curve.
  - **Deep Traps:** Characterized by a larger energy gap, these traps necessitate higher temperatures to free the trapped carriers. They correspond to high-temperature glow peaks; extremely deep traps may not be emptied during standard readout cycles.
- **Recombination Centers:** Alongside traps, the material contains luminescent centers. When a released electron encounters a hole at one of these centers, recombination occurs, accompanied by the emission of a photon. These centers are typically linked to the intentionally introduced dopant ions (e.g., Ti in LiF or Dy in $CaSO_4$).

3.2.3 Trap Population and Filling

During exposure to ionizing radiation, a competition exists between the trapping of charge carriers and their immediate recombination. The extent to which these traps are filled depends on parameters like trap density, capture cross-sections, and ambient temperature. Crucially, up to the point of saturation, the population of filled traps maintains a proportional relationship with the absorbed radiation dose.

3.2.4 Kinetic Model and Glow Curve Formalism

For first-order thermoluminescence kinetics, the TL intensity can be represented as

$$I(T) = -\frac{dn}{dt} = ns\exp\left(-\frac{E}{kT}\right), \tag{1}$$

where n is the trapped carrier population, s is the frequency factor, E is trap depth (activation energy), k is the Boltzmann constant, and T is absolute temperature. Under a linear heating rate $\beta = dT/dt$, the peak position shifts with E, s, and β. The integrated light output,

$$L = \int I(T)\,dt, \tag{2}$$

is proportional to absorbed dose over the linear operating region and forms the basis of dose calibration [1, 2].

3.2.5 Fading

Fading refers to the spontaneous loss of the stored dosimetric signal over time. Even at room temperature, shallow traps may undergo thermal emptying, leading to a reduction in the measured dose if analysis is delayed.

3.3 Heating, Recombination, and Light Emission

3.3.1 Controlled Heating / Thermal Stimulation

During the readout phase, the TLD is subjected to a controlled linear heating profile (e.g., 4°C/s). As the temperature escalates, the thermal energy (kT) becomes adequate to provide the necessary activation energy for electrons residing in traps of specific depths, elevating them into the conduction band. The likelihood of an electron escaping is exponentially dependent on both the temperature and the trap's activation energy. Consequently, distinct sets of traps release their electrons at characteristic temperatures.

3.3.2 Migration and Recombination

Once liberated into the conduction band, the electrons become mobile. They migrate through the crystal lattice until they interact with a recombination center. The ensuing recombination process releases excess energy as a visible photon. This luminescent output is captured by a PMT, and the recorded light intensity is plotted as a function of temperature.

3.3.3 Glow Curve Peaks

The varying depths of different traps mean that electrons are released across a range of temperatures, resulting in a glow curve characterized by multiple distinct peaks. Each peak signifies the depopulation of a specific trap type. The area under a given peak correlates with the number of released electrons, which ideally reflects the dose absorbed by that specific trap set. The peak's temperature location provides insight into the trap's activation energy. For dosimetric purposes, materials that yield well-separated peaks are highly desirable.

3.3.4 Example and Stability of Peaks

For practical dosimetry, a stable peak is selected. For instance, in $CaSO_4$:Dy, the primary dosimetric peak occurs at roughly 280°C. This temperature is sufficient to prevent rapid room-temperature fading but low enough to avoid complications associated with extreme heating. Conversely, low-temperature peaks are susceptible to rapid fading and are often intentionally erased during a pre-heat cycle before the final reading. This reported peak position for $CaSO_4$:Dy is consistent with experimental literature where the dosimetric peak appears near 277.5°C in low-dose studies [4].

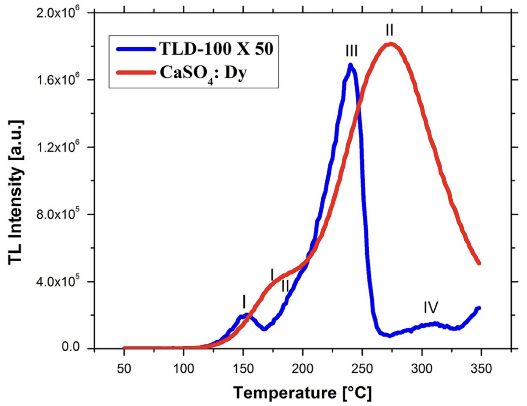

Fig. 1. Glow curve at a dose of 14.6 mGy for TLD-100 and $CaSO_4$:Dy. Note that the thermoluminescent intensity of TLD-100 is multiplied by 50.

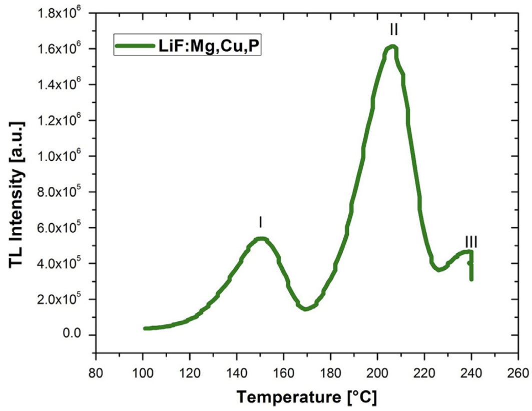

Fig. 2. Glow curve at a dose of 14.6 mGy for LiF:Mg, Cu, P.

3.4 Annealing

Following the readout, a small fraction of deeply trapped electrons may remain. To clear these residual signals and restore the material's structural equilibrium for reuse, an annealing procedure is necessary. This involves heating the TLD phosphors in a specialized oven—typically at 400°C for one hour—ensuring that all traps are completely emptied without degrading the material's properties. Controlled re-annealing also affects sensitivity and glow-curve characteristics, so a fixed and reproducible annealing protocol is essential for calibration consistency [5, 3].

3.5 Working Principle of a TLD Personnel Monitoring Badge Reader

The readout process involves several sequential steps:

1. Exposure to radiation causes electrons in the TLD material to be elevated and captured in metastable traps.
2. The quantity of these trapped electrons directly corresponds to the received radiation dose.
3. The exposed badge is inserted into the TLD reader.
4. A heating mechanism applies a precisely controlled thermal cycle to the TLD elements.
5. The increasing temperature liberates the trapped electrons.
6. As these electrons undergo recombination, they emit energy as visible light.
7. A Photomultiplier Tube (PMT) detects this emitted luminescence.
8. The PMT converts the optical signal into an equivalent electrical current.
9. This current is amplified and processed by the reader's internal electronics.
10. Finally, after applying pre-determined calibration factors, the system displays the calculated absorbed dose.

In clinical quality assurance workflows, this sequence is supplemented by routine checks for element correction coefficients, reader stability, background subtraction, and batch-wise calibration verification to maintain measurement uncertainty within acceptable limits [3, 6].

3.6 Reason for Using Three TLD Phosphors in a Personnel Monitoring Badge

The incorporation of three distinct TLD elements within a single badge is a deliberate design choice aimed at enhancing measurement accuracy:

- **Radiation Type Discrimination:** The unique filters (metal, plastic, open window) over each element interact differently with X-rays, gamma, and beta radiation. Analyzing the relative responses of the three elements allows for the identification of the incident radiation type.

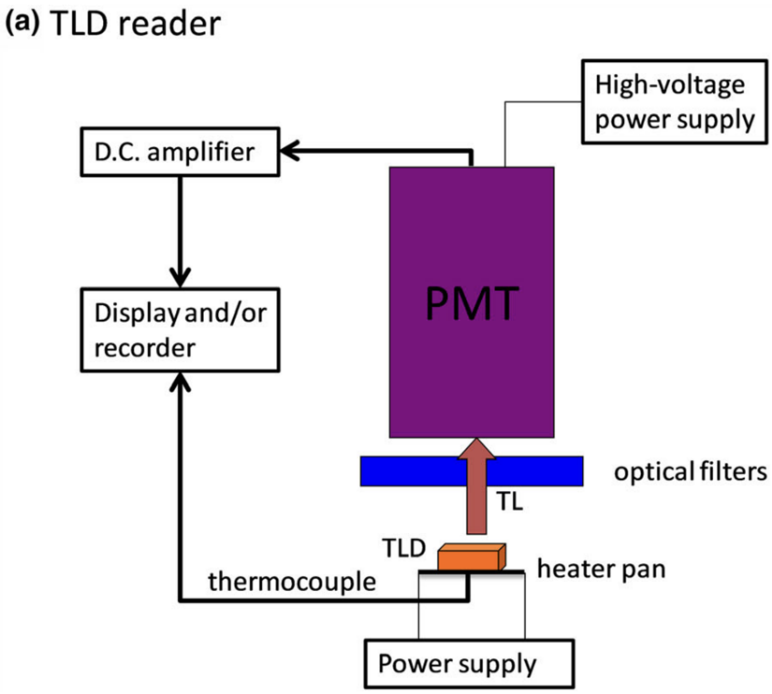

Figure 3: Schematic representation of a TLD reader system.

- **Energy Dependence Compensation:** Materials like $CaSO_4$:Dy exhibit significant variations in response depending on the photon energy. The differential attenuation provided by the filters enables the application of correction factors, mitigating this energy dependence.
- **Separation of Beta and Gamma Components:** Beta particles are largely stopped by the metal and plastic filters but easily penetrate the open window. Comparing the readouts allows the system to isolate and quantify the beta dose separately from the gamma dose.

3.7 Dose, Temperature, and Energy Dependence of TL Intensity

- **Dose Dependence:** Within the standard occupational exposure range, the TL intensity maintains a linear relationship with the absorbed dose, a critical feature for reliable dosimetry. At extremely high doses, the material may exhibit supralinearity and eventual saturation as deeper traps become fully occupied.
- **Temperature Dependence:** The readout is critically dependent on the heating profile. Insufficient heating fails to release all trapped charges, causing an underestimation of the dose. Conversely, excessive temperatures can induce thermal quenching, degrading the luminescent signal. Thus, precise thermal control is paramount.
- **Energy Dependence:** The efficiency of the TL response varies with the energy of the incoming radiation. Typically, low-energy photons yield a higher TL response per unit dose than high-energy photons. The badge's filter system and specific calibration algorithms are employed to correct this intrinsic material behavior.

These dependencies are not only theoretical; they directly influence calibration and uncertainty budgets in medical dosimetry. Therefore, calibration should ideally be performed under beam qualities and geometries close to actual use conditions [3, 1].

3.8 Advantages of Thermoluminescent Dosimeters (TLD)

- **Wide Dose Range:** Capable of measuring exposures from micrograys up to several grays.
- **High Sensitivity:** Excellent at detecting minimal radiation levels.

- **Reusability:** Following proper thermal annealing, the dosimeters can be redeployed multiple times.
- **Tissue Equivalence:** Specific phosphors (like LiF:Mg,Ti) closely match the effective atomic number of human tissue.
- **Passive Operation:** They require no electrical power during the exposure period.
- **Compact Form Factor:** Their small size and robust nature make them ideal for daily personnel wear.
- **Linearity:** They offer a highly linear response across the typical occupational dose range.

3.9 Limitations of Thermoluminescent Dosimeters (TLD)

- **Destructive Readout:** The process of reading the dose erases the stored signal, preventing a direct re-read of the exposure.
- **Fading:** The trapped signal can spontaneously degrade over time, particularly in warm environments.
- **Complex Evaluation:** Extracting the dose requires a sophisticated reader unit and a tightly controlled heating cycle.
- **Energy Dependence:** Certain highly sensitive materials require complex filtration to correct for varying responses to different photon energies.
- **Delayed Results:** Being a passive device, it does not offer real-time or instantaneous dose rate information.

3.10 Heating Mechanism of the TLD Badge Reader

The integrity of the TLD readout heavily relies on the reader's heating system:

- The reader elevates the temperature of the TLD discs to a precisely targeted level (often around 285°C) to stimulate electron release.
- Modern automated readers frequently utilize a hot nitrogen gas system. A dedicated heater warms the nitrogen, which then flows uniformly over the discs, ensuring efficient and consistent thermal transfer.
- The system includes a heating element (e.g., nichrome wire) coupled with advanced temperature control electronics to achieve a rapid ramp-up and a stable hold at the target temperature.
- Thermocouples or similar sensors continuously monitor the temperature, feeding data back to the control system to maintain a precise thermal profile.
- This highly regulated heating process guarantees reproducible glow curves, which is the foundational requirement for accurate dose calculation.

3.11 Suggested Image Placement in Theory Section

For improving clarity and presentation quality, include the following figures at these points in this chapter:

- After the subsection *Detection Material*: Insert the badge photograph/schematic (recommended file: Figures/TLD_Badge.jpg) to show cassette, card, and filter arrangement.
- After *Basic Insulator Band Model*: Insert an energy-band/trap diagram (create as Figures/energy_band_diagram.png) showing valence band, conduction band, traps, and recombination centers.
- After *Glow Curve Peaks*: Insert a representative glow curve plot (create as Figures/glow_curve_example.png) with labeled dosimetric peak near 280°C for $CaSO_4$:Dy.
- After *Dose, Temperature, and Energy Dependence*: Insert the calibration response figure (existing file: Figures/calibration_curve.png) to support the linearity discussion.

4 Observation

We brought the TLD badges to the medical X-ray collimator set up, where we set the focus to film distance 100 cm. We noted down the badge no and irradiated the badge with x ray keeping the KVp = 100 V and irradiation time 0.1 sec.

- FTD = 90 cm
- FFD = 100 cm
- Field area = 20×20 cm^2
- KVp = 100 V
- Time = 0.1 sec

We then irradiated two badges with 63 mA and 125 mA and tried to find the unknown dose to both TLD badges. While taking the reading in badge reader, the inlet pressure of the gas was adjusted to 2 Kg/cm^2.

4.1 Observation table of Data

Table 4: Experimental data

Currents (mA)	TLD ID	Xray Dose (mGy)	D1 – Int TL	D2 – Int TL	D3 – Int TL	Avg Dose (mGy)
	108	0.26	421	2643	2490	
20	153	0.26	359	2653	2574	0.26
	143	0.26	373	2653	2477	
40	237	0.46	524	4582	4661	0.46
	183	0.96	945	9552	9715	
80	229	0.9	937	8825	9146	0.93
	146	1.8	1761	18495	17541	
160	182	1.81	1703	17652	18565	1.805
	248	2.21	2087	21607	22560	
200	147	2.3	2127	22725	23366	2.255
	235 (Background)	0	229	479	496	
	247 (Background)	0	176	196	259	0

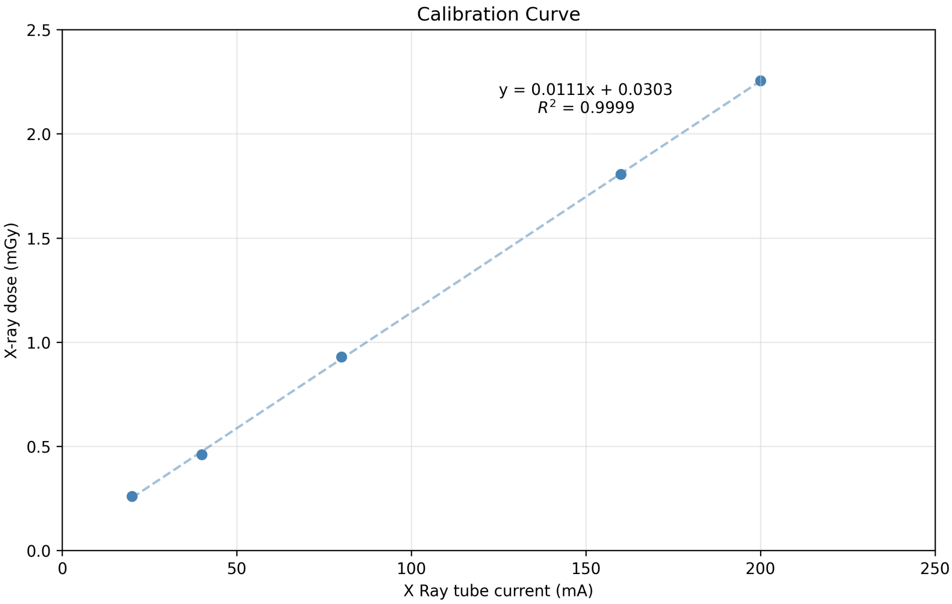

Figure 4: Calibration Curve

The equation of the curve derived from the experimental plot is:

$$y = 0.0111x + 0.0303 \tag{3}$$

This follows the linear form $y = mx + c$, where:

- y is the X-ray dose (mGy)
- x is the X-ray tube current (mA)

So the linearity of the dose with tube current is checked.

5 Calculation

Using equation (1), the absorbed dose of TLD card 146 (unknown 1) due to exposure for 63 mA current is given by 0.7296 mGy.

Using equation (1), the absorbed dose of TLD card 108 (unknown 2) due to exposure at 125 mA current is given by 1.4178 mGy.

5.1 Error Calculation

According to the TLD badge reader, absorbed dose of TLD 189 (unknown 1) and TLD card 134 (unknown 2) are given by 0.74 mGy and 1.4 mGy respectively. So, the corresponding errors are 1.4% and 1.2%.

6 Conclusion

The calibration of the TLD personnel monitoring badge reader was successfully executed utilizing a medical X-ray unit as the primary radiation source. Throughout the experiment, a distinct linear correlation was established between the delivered X-ray dose and the applied tube current. This finding confirms the anticipated linear dose-response behavior of the $CaSO_4$:Dy TLD badges within the evaluated dose range, reinforcing their efficacy and appropriateness for precise occupational personnel dosimetry.

Furthermore, the derived calibration curve was effectively employed to deduce the radiation exposure of unknown samples. The mathematically calculated dose values exhibited excellent agreement with the automated readings provided by the TLD badge reader's software. Specifically, the remarkably low percentage errors obtained (1.4% and 1.2%) underscore the high repeatability, inherent reliability, and exceptional precision of the TLD badge reader system in evaluating unknown radiation doses.

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